SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code: CSA17	**Course Title:** Artificial Intelligence
CO Assessed: CO1 – Apply (BL3)	**Assessment:** Assessment Tool 1 – Analytical Problem Solving
Weightage: 35% of CIA	**Duration / Marks:** 60 mins
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Section / Batch: _____	**Date:** 11-10-2026

CO1 Statement: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

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Signature of the Student: _____

Annexure A – Analytical Problem Solving: Answers

Every answer states its assumptions, the method and why it was chosen, the working, and an interpretation. Units: marks, minutes, percentage points (pp), score units and expected utility.

Question 1: Two-Room Vacuum World

(a) PEAS description [4 Marks]

Method: PEAS (Performance measure, Environment, Actuators, Sensors) is the standard way to specify a task environment before designing an agent, so each element is taken directly from the data given.

Element	Description
Performance measure	Over a lifetime of T = 6 time steps: +1 for every clean square at the end of each step (maximum 2 per step, 12 in all) and a penalty of p for every Left or Right move (p = 1 unless told otherwise).
Environment	Two squares A and B, each Dirty or Clean; both start Dirty and the agent starts in A. Once a square is cleaned it stays clean (dirt never reappears). Discrete, deterministic, static, single-agent, sequential; only the current square is observed (partially observable).
Actuators	Suck (clean the current square), Left, Right (move to the other square), NoOp (do nothing).
Sensors	A location sensor and a dirt sensor for the current square, giving the percept [location, Dirty/Clean].

(b) Agent X (simple reflex), p = 1 [5 Marks]

Assumptions: the score of a step is (clean squares at the end of the step) minus the penalty for that step's move; the total is the sum over t = 1 to 6. **Method:** step-by-step simulation of the rule "if Dirty, Suck; otherwise move to the other square".

t	Position, percept	Action	Clean squares at end of step	Move penalty (p = 1)	Step score
1	A, Dirty	Suck	1 (A)	0	+1
2	A, Clean	Right	1 (A)	1	0
3	B, Dirty	Suck	2 (A, B)	0	+2
4	B, Clean	Left	2	1	+1
5	A, Clean	Right	2	1	+1
6	B, Clean	Left	2	1	+1
Total			**10**	**4**	**6**

Total score of X (p = 1) = 10 - 4 = 6. After both squares are clean, X keeps moving back and forth and pays a penalty on every one of the last three steps.

(c) Agent Y (reflex with internal state), p = 1 [4 Marks]

Method: same simulation, but Y stores which squares it has cleaned. After step 3 the state is "A and B both cleaned", so the rule gives NoOp.

t	Position, percept	Internal state (cleaned)	Action	Clean squares	Penalty	Step score
1	A, Dirty	{}	Suck	1	0	+1
2	A, Clean	{A}	Right	1	1	0

t	Position, percept	Internal state (cleaned)	Action	Clean squares	Penalty	Step score
3	B, Dirty	{A}	Suck	2	0	+2
4	B, Clean	{A, B}	NoOp	2	0	+2
5	B, Clean	{A, B}	NoOp	2	0	+2
6	B, Clean	{A, B}	NoOp	2	0	+2
Total				**10**	**1**	**9**

Total score of Y (p = 1) = 10 - 1 = 9.

(d) Scores as functions of p [3 Marks]

Method: both agents have the same clean-square total (1 + 1 + 2 + 2 + 2 + 2 = 10); only the number of moves differs (X moves 4 times: t = 2, 4, 5, 6; Y moves once: t = 2).

Agent	Score as a function of p	p = 0	p = 1	p = 2
X	$S_X(p) = 10 - 4p$	10	6	2
Y	$S_Y(p) = 10 - p$	10	9	8
Difference	$S_Y - S_X = 3p$	0	3	6

$S_Y > S_X$ exactly when $3p > 0$, i.e. **for every p > 0**. For p = 0 the two tie (10 = 10); for p < 0 (moving would be rewarded) X would be better.

(e) Which agent is rational? [4 Marks]

Y is the rational agent (for p > 0). Checking every possible sequence of the four actions over the 6 steps gives a best possible score of 10 - p, which Y achieves (at least one move is unavoidable to reach B). X scores 10 - 4p, which is lower whenever p > 0.

Interpretation. Rationality is defined by the performance measure, the agent's prior knowledge, its available actions and its *percept sequence*. The measure charges p for every move, so moving after everything is clean is wasteful. X and Y receive identical percepts at each step, but X's agent function depends only on the *current* percept. The percept [A, Clean] at t = 2 needs the action Right (B is still dirty), while the identical percept [A, Clean] at t = 5 needs NoOp. No rule based on the current percept alone can give both answers; Y can, because its internal state records the percept history. X therefore cannot match Y for p > 0, however it is tuned. If p were 0, X would be equally rational, as the wasted moves would cost nothing.

Question 2: Task Environment Analysis and Agent Selection

(a) Classification of E1, E2, E3 [9 Marks]

Assumptions: E1 = taxi in city traffic; E2 = camera-guided part-picking robot on a moving conveyor (each part is judged on its own); E3 = chess game with a clock. **Method:** classify each task on the six standard properties and score the harder value 1, the easier 0 (semi-dynamic 0.5).

Property	E1: Taxi	E2: Part-picking robot	E3: Chess with clock
Observability	Partially (1) – cannot see around corners, other drivers' intent, road surface	Partially (1) – camera noise/occlusion, defects may be hard to see	Fully (0) – whole board visible
Number of agents	Multi-agent (1) – other vehicles, pedestrians	Single (0) – only the robot acts on the belt	Multi-agent (1) – an opponent
Determinism	Stochastic (1) – traffic, signals, accidents	Stochastic (1) – slip/grasp errors, uncertain defect pattern	Deterministic (0) – no chance element
Episodic / Sequential	Sequential (1) – each manoeuvre shapes later ones	Episodic (0) – each part's decision is independent of earlier ones	Sequential (1) – each move affects later moves
Static / Dynamic	Dynamic (1) – world moves while the taxi thinks	Dynamic (1) – belt keeps moving	Semi-dynamic (0.5) – board is unchanged but the clock penalises time
Discrete / Continuous	Continuous (1) – positions, speeds, steering	Continuous (1) – positions and arm motion	Discrete (0) – finite positions and moves

(b) Difficulty scores and ranking [4 Marks]

Environment	Obs.	Agents	Determ.	Epis./Seq.	Static/Dyn.	Disc./Cont.	Total
E1: Taxi	1	1	1	1	1	1	**6**
E2: Part-picking robot	1	0	1	0	1	1	**4**
E3: Chess with clock	0	1	0	1	0.5	0	**2.5**

Ranking, hardest to easiest: E1 (6) > E2 (4) > E3 (2.5).

(c) Agent architecture for the hardest and the easiest environment [4 Marks]

Environment	Suitable architecture	Justification from the classification
E1 (hardest, score 6)	**Utility-based agent** (with an internal model and a learning component)	Partially observable, so it needs an internal state/model of hidden traffic. Stochastic, dynamic and multi-agent, so outcomes are uncertain and it must compare options by *expected utility*. A goal ("reach the airport") cannot separate a fast, risky route from a slower, safe, comfortable one, and a simple reflex or goal-only agent cannot trade these off. Learning helps with changing traffic and rare events.
E3 (easiest, score 2.5)	**Goal-based agent** (game-tree search with an evaluation function)	Fully observable and deterministic, so the board state is known exactly and the effect of each move can be predicted without an internal model of hidden state. It is sequential, so a simple reflex rule is not enough and the agent must look ahead to reach checkmate. The only extra complication is the opponent and the clock; a depth-limited evaluation function suffices.

(d) Interpretation [3 Marks]

The complexity of the agent program is set by the *properties* of the task environment, not by its domain label. Partial observability forces the agent to keep and update an internal state; stochasticity and multiple agents force it to reason with probabilities and expected utility instead of fixed plans; a sequential task needs look-ahead because actions have lasting effects; a dynamic task needs fast, anytime decisions; a continuous state needs approximation. Both chess and driving are "hard" for humans, yet chess scores 2.5 and the taxi 6, and E2 (a physical robot) is easier than E1 as each part is an independent (episodic) decision. So a simple reflex agent suffices only when the task is fully observable and episodic, and each extra hard property pushes the design towards model-based, goal-based and utility-based programs.

Question 3: Turing Test Data Across the History of AI

(a) Correct-identification rate and fooling rate [6 Marks]

Method: correct-identification rate = correct / 50; fooling rate = 1 - rate = (50 - correct) / 50, since every interrogation was either a correct identification or the machine being taken for a human.

System	Correct / 50	Correct-identification rate	Machine taken for human	Fooling rate
ELIZA (1960s)	47	47 / 50 = 94%	3	3 / 50 = **6%**
Chatbot B (2010s)	36	36 / 50 = 72%	14	14 / 50 = **28%**
System C (2020s)	29	29 / 50 = 58%	21	21 / 50 = **42%**

(b) Turing's 70% criterion and distance from chance [5 Marks]

Method. (1) Compare each correct-identification rate with Turing's threshold of 70% ("no more than 70%" correct, i.e. a fooling rate of at least 30%). (2) Measure the distance from the 50% guessing level in percentage points, and test it with a one-sample proportion z-test, z = (rate - 0.5) / SE, where SE = sqrt(0.5 x 0.5 / 50) = 0.0707. At the 5% level, |z| > 1.96 means the judges did better than guessing.

System	Rate	Meets 70% criterion (rate <= 70%)?	Distance from 50%	z	Two-sided p	Different from guessing?
ELIZA	94%	No (24 pp above 70%)	+44 pp	6.22	< 0.001	Yes – easily identified
Chatbot B	72%	No, narrowly (2 pp above 70%)	+22 pp	3.11	0.002	Yes
System C	58%	**Yes** (12 pp below 70%)	+8 pp	1.13	0.26	No – cannot be told apart from guessing

Result: only **System C** satisfies the 70% criterion. Chatbot B misses it by 2 percentage points; that gap is not statistically significant (z = 0.31 against 70%), so B is borderline, not clearly failing. ELIZA is far from the criterion and 44 pp above chance. Only System C is statistically indistinguishable from the 50% chance level (p = 0.26 with only 50 interrogations, so the evidence is limited).

(c) Improvement in fooling rate [5 Marks]

Comparison	Fooling rates	Improvement (percentage points)	Improvement as a ratio
ELIZA to System C	6% to 42%	42 - 6 = **+36 pp**	42 / 6 = **7.0 times** (an increase of 600%)
Chatbot B to System C	28% to 42%	42 - 28 = **+14 pp**	42 / 28 = **1.5 times** (an increase of 50%)

Interpretation: percentage points measure the absolute change, ratios the relative one. The ratio exaggerates progress from ELIZA because it starts from a very low base (6%), while the +14 pp step from Chatbot B (statistical NLP) to the neural language model System C is the gain that finally brings the machine within the 70% criterion.

(d) Does a high fooling rate prove thinking or rational acting? [4 Marks]

The four approaches to defining AI are: thinking humanly, thinking rationally, acting humanly and acting rationally. The Turing Test measures only **acting humanly**: whether the *text behaviour* of the machine can be told from a person's in a short conversation. A 42% fooling rate therefore shows convincing imitation, not that System C thinks or acts rationally:

Reason	Explanation
Chinese Room argument (Searle)	A system can manipulate symbols so well that its replies look understanding, yet it may have no understanding of what the symbols mean. Passing the test does not show thinking.
Acting rationally is a different target	Rational action means choosing the action that maximises expected performance. Imitating a human, including human slips, is not the same as achieving the best outcome, and humans are not always rational.
Weak evidence	The result depends on the five-minute limit, the topic, the judges and a small sample (n = 50); it says nothing about reasoning outside the conversation.

Question 4: Rational Route Choice under Uncertainty

Assumptions: *utility U = 100 - T - 20L, with L = 1 if T > 55 min and L = 0 otherwise; disruptions occur with the stated probability and give the stated time.* ***Method:*** *a rational agent picks the action with the highest expected utility, EU = sum of probability x utility over the outcomes.*

(a) Utility of each route in each outcome [4 Marks]

Route	Outcome	T (min)	On time? (T <= 55)	L	U = 100 - T - 20L
R1 Highway	Normal	30	Yes	0	100 - 30 - 0 = **70**
R1 Highway	Disrupted	70	**LATE**	1	100 - 70 - 20 = **10**
R2 City road	Normal	45	Yes	0	100 - 45 - 0 = **55**
R2 City road	Disrupted	60	**LATE**	1	100 - 60 - 20 = **20**
R3 Ring road	Normal / never disrupted	50	Yes	0	100 - 50 - 0 = **50**

(b) Expected utilities and the rational choice [5 Marks]

Route	Expected utility	Value
R1	0.70 x 70 + 0.30 x 10 = 49 + 3	**52.0**
R2	0.90 x 55 + 0.10 x 20 = 49.5 + 2	**51.5**
R3	1.00 x 50	**50.0**

The rational agent should choose R1 (Highway), EU = 52.0. The margin is small (0.5 over R2). R1 stays best only while its disruption probability q < 0.308: EU(R1) = 70 - 60q equals EU(R2) = 51.5 at q = 0.308 and EU(R3) = 50 at q = 0.333.

(c) New percept: radio alert, P(disruption on R1) = 0.80 [4 Marks]

EU(R1) = 0.20 x 70 + 0.80 x 10 = 14 + 8 = **22.0**. R2 (51.5) and R3 (50) are unchanged, because the alert says nothing about them.

Route	EU before alert	EU after alert
R1	52.0	22.0
R2	51.5	**51.5 (best)**
R3	50.0	50.0

New decision: R2 (City road), EU = 51.5. R1 has dropped from first to last.

(d) The choice from (b) and the disruption occurs [4 Marks]

The agent chose R1 and the accident happened, so it arrives at T = 70 min, late: **U = 100 - 70 - 20 = 10**, 42 below its expected 52 and 40 below what R3 would have given with certainty.
The agent was not irrational. When it decided, R1 had the highest expected utility given everything it knew (30% chance of a disruption). A bad outcome from a sound decision is bad luck, not irrationality.

	Rational agent	Omniscient agent
Knows	Its percept sequence, prior knowledge and the probabilities	The actual outcome of every action in advance
Chooses	The action with maximum *expected* utility	The action with the best *actual* utility (here it would avoid R1, e.g. R3 with 50, or R2 with 55 if its signals work)
Can be built?	Yes	No – impossible in an uncertain world
Can it be disappointed?	Yes, outcomes can be worse than expected	Never

(e) Interpretation [3 Marks]

Rationality is relative to the **performance measure** (here U, which trades time against a 20-point lateness penalty; a larger penalty or a different weight would change the ranking) and to the **percept sequence** (what the agent has perceived so far). The three roads did not change between (b) and (c); what changed is the agent's information. The radio alert raised its belief about the highway's disruption from 0.30 to 0.80, which lowered EU(R1) from 52 to 22 and made R2 the best choice. Using new percepts to update beliefs, and then choosing by expected utility, is exactly what makes the agent rational; ignoring the alert and staying on R1 would have been irrational.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Evaluation Page (for faculty use)

Student: Deenesh Ra S T Reg. No.: 192321149

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge

Signature: _____
Date: _____

Course Coordinator

Signature: _____
Date: _____

Program Director

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Date: _____